Tracking efficiencies from $J/\psi \to \mu^+\mu^-$ and hadronic corrections

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Motivation and strategy

Why do we care?

- Imperfect modelling of track reconstruction efficiency in MC
- Physics analyses that rely on efficiencies from MC must correct for this by reweighting

What do we provide?

1. Tracking efficiency ratio between data/MC of muon long tracks
   - Long track efficiency from product of Velo and SciFi efficiencies
   - 2D p–η binning
   - Dominant systematic uncertainty from MuonUT-Combined difference
2. Hadronic corrections for pions, kaons and protons
   - Mainly due to uncertainty in material budget
   - Using binning in η, with the same bin edges as for muons

Part I: Tracking efficiencies with $J/\psi \to \mu^+\mu^-$

Part II: Hadronic corrections

Part I: Tracking efficiencies with $J/\psi \to \mu^+\mu^-$

Part II: Hadronic corrections

Tag-and-probe method

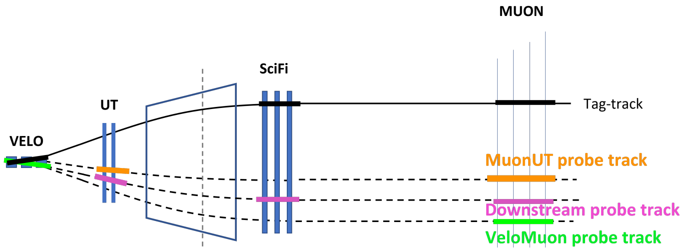


Figure by Rowina

- Tag-and-probe method with $J/\psi \to \mu^+\mu^-$
- Fully reconstruct one muon...
- ... and partially reconstruct the other muon
- Match hits in specific sub-detector with partially reconstructed track

$$\epsilon_{\text{track}} = \frac{N_{\text{matched}}}{N_{\text{matched}} + N_{\text{failed}}}$$

Strategy for tracking efficiency determination:

- Combine efficiencies of two methods:
  - VeloMuon: SciFi efficiency
  - Downstream: Velo efficiency
  - Combined: Velo $\times$ SciFi efficiencies
- Cross check with long track efficiency from the MuonUT method
  - Main systematic

Overview of previous work

Long history of previous work

- Method and results carefully documented in Rowina's thesis here
- Latest presentations in RTA-WP4 on
  - muon tracking efficiencies here
  - hadronic corrections here

Tracking efficiencies are determined using TrackCalib2

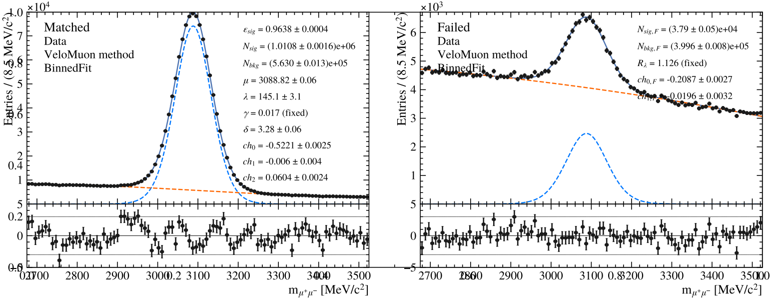
- GitLab repository here
- Mass fit done on tuples from AP to extract efficiencies
- Note: Analysts should not have to run TrackCalib2 themselves, unless the released numbers are not suitable (need different binning, etc)

Datasets considered

Datasets considered:

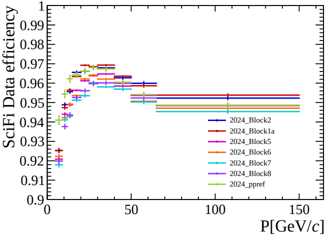
- 2024
  - Analysis of blocks 2, 1, 5, 6, 7, 8, *pp* ref mostly done
  - Currently waiting for large MC production to finish
  - Blocks 4 and 3 don't have MC yet
- 2025
  - Analysis of pre-TS data in progress
  - Will request 2025 post-TS MC soon
- 2026
  - Plan to submit AP and check data in the next week or two
  - Full analysis will follow after 2025 results are completed

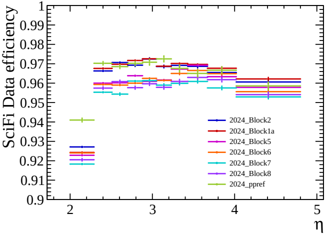
Example of fit to extract efficiency (2024 block 5, 1D η bin 3)

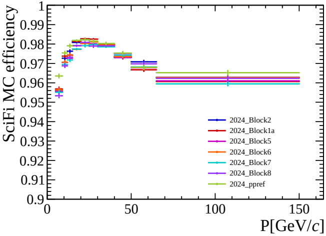


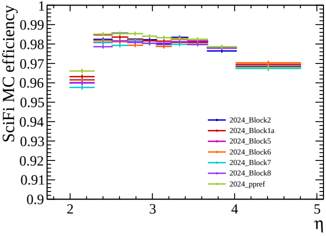
- Efficiency is a free parameter in the fit
- Matched and failed samples simultaneously fitted to ensure correct uncertainties

Comparison of SciFi tracking efficiencies in 2024 data blocks



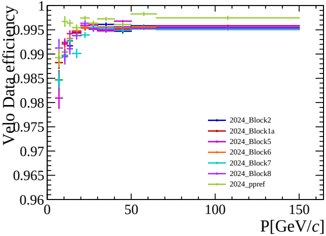


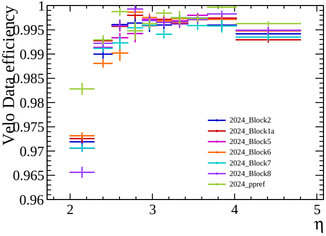


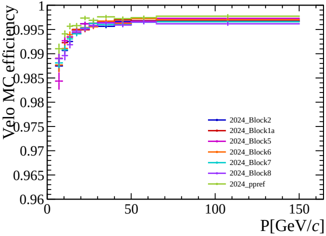


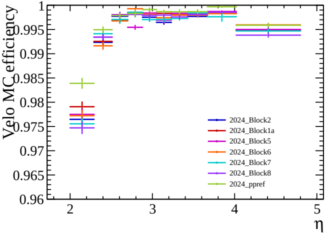
Efficiency comparisons 2024

Comparison of Velo tracking efficiencies in 2024 data blocks









General observations for 2024 results

General observations on tracking efficiencies:

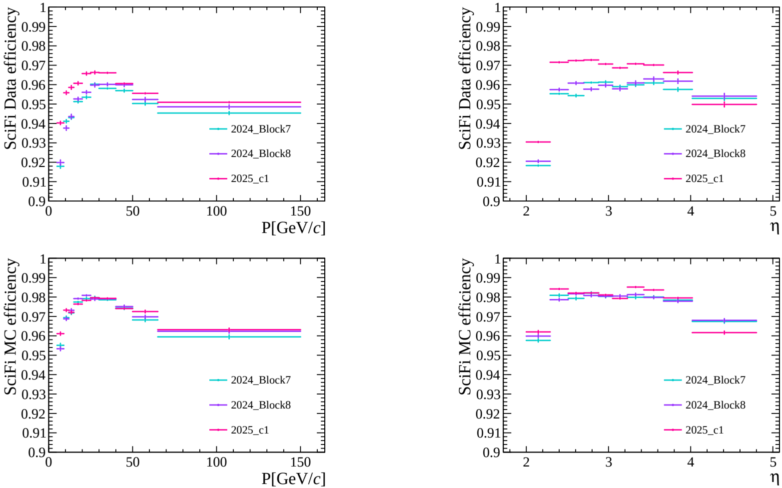
1. All data sets show consistent trends
2. Performance seems to (slowly) drop over time

This is very encouraging! Several issues resolved, including:

1. Unusually high efficiencies in blocks 7/8
2. Missing binds when rerunning reconstruction
3. Large charge asymmetries in MuonUT samples

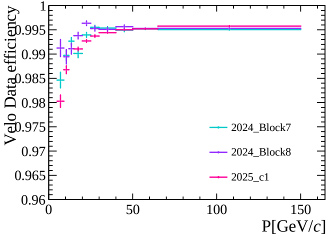
Efficiency comparisons 2025

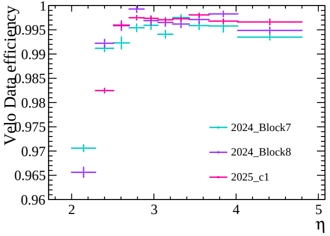
Comparison of SciFi tracking efficiencies in 2025, compared with 2024

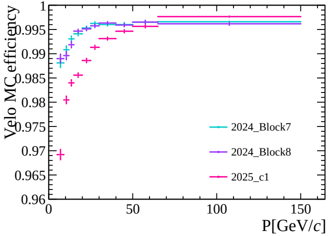


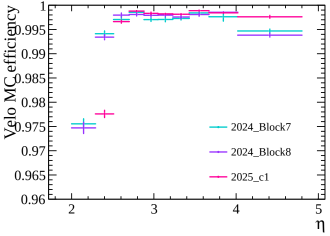
Efficiency comparisons 2025

Comparison of Velo tracking efficiencies in 2025, compared with 2024

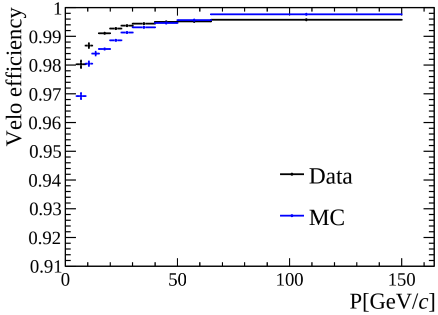


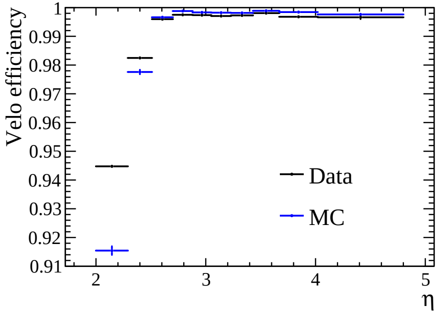






2025 Data-MC discrepancy





Need to understand why MC underestimates the Velo efficiency in 2025!

General observations for 2025 results

General observations on tracking efficiencies:

- SciFi efficiencies slightly higher in data, compared to 2024
- Velo MC efficiencies greatly underestimated, work in progress to understand why
- In terms of timescale, once the Velo efficiencies are understood and more MC is requested, it should be very quick to produce correction tables for 2025 as well

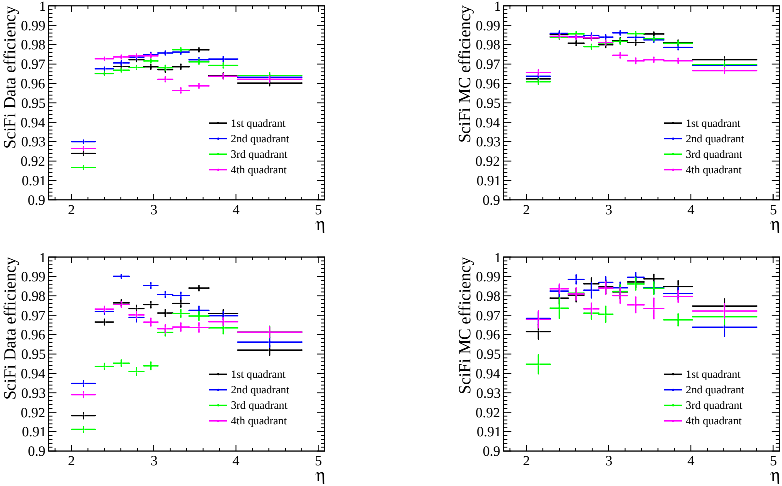
Block 1a vs 1b

Do we need to split Block 1 into Block 1a and Block 1b?

- SciFi T2L0Q0M1H0 FE was excluded on 17th August
  - Block 1a: Fill 9982–10012
  - Block 1b: Fill 10017–10056
- Integrated luminosities are 0.92 fb^{-1} and 0.31 fb^{-1}, respectively
- SciFi effectively has a hole in the 3rd quadrant

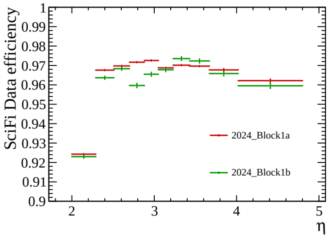
Block 1a vs 1b

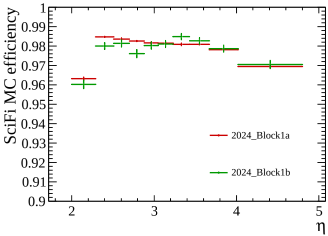
Block 1a (top) vs 1b (bottom) for data (left) and MC (right)



Block 1a vs 1b

Integrated over all quadrants, for data (left) and MC (right)





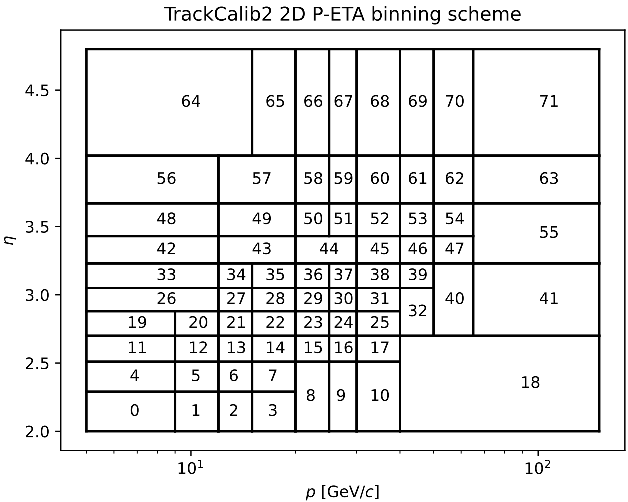
- SciFi "hole" is visible in the 3rd quadrant in data, but not MC
- The integrated numbers show reasonable agreement with MC, with a small drop in efficiency for a single η bin
- For now, tracking efficiencies are only provided for block 1a...
- ... but separate MC might be necessary for block 1b in case a separate calibration table is wanted

2D binning

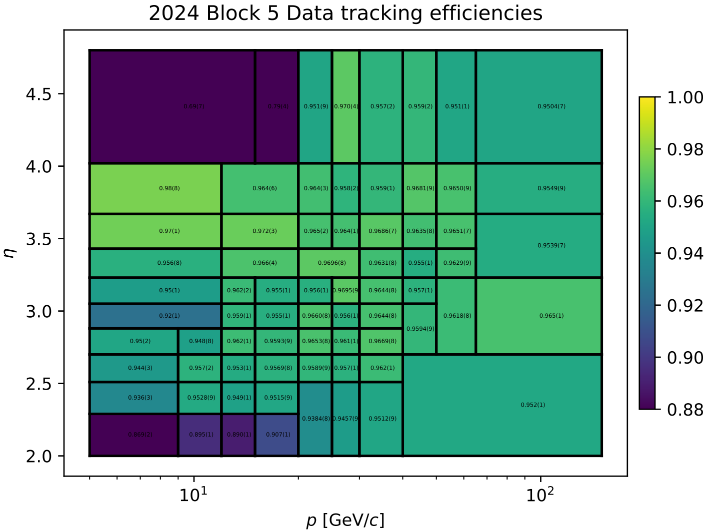
- In 1D we have 10 bins in p and η...
- ... 10×10 2D bins is impossible
  - Insufficient statistics at high (low) p and low (high) η
  - But most bins <u>do</u> have sufficient statistics
- Strategy: Start with 10×10 bins and merge bins
  - Only consider MC, which is important for constraining the fit shape
  - Results driven by SciFi efficiencies, so only consider VeloMuon yields
  - Try to merge bins along η at high p and vice versa
- The next plots contain many (tiny) numbers, apologies for this!

2D binning

Final 2D binning in p vs η with 72 bins:

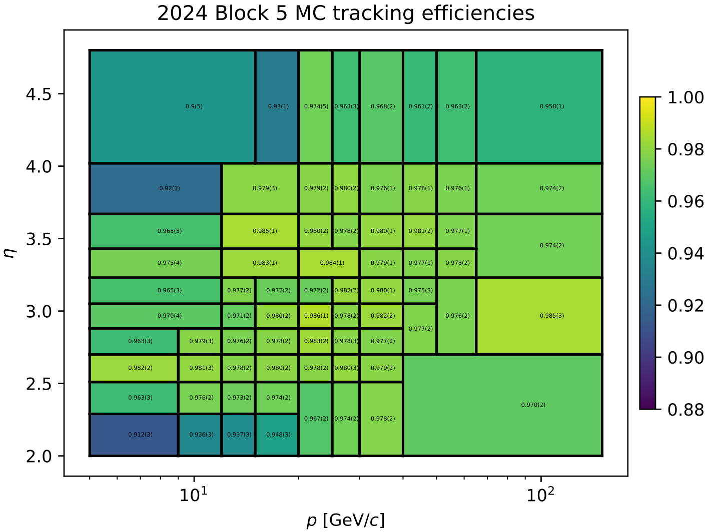


2D binning



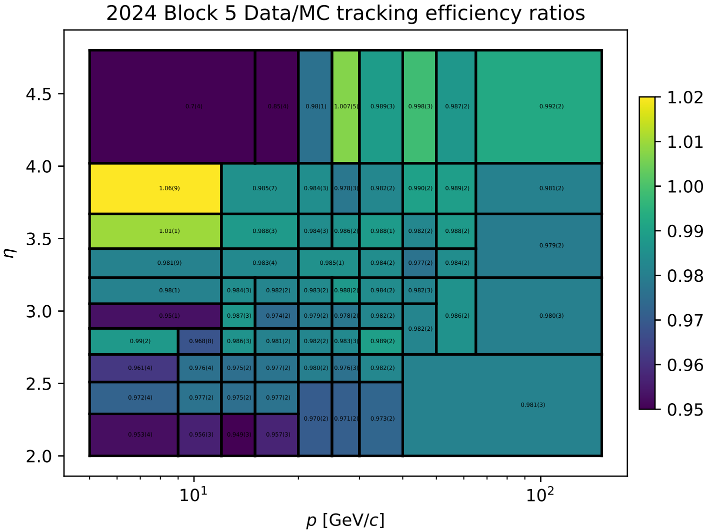
2024 Block 5 Data tracking efficiencies
0.69(7)
0.79(4)
0.951(9)
0.970(4)
0.957(2)
0.959(2)
0.951(1)
0.9504(7)
0.98(8)
0.964(6)
0.964(3)
0.958(2)
0.959(1)
0.9681(9)
0.9650(9)
0.9549(9)
0.97(1)
0.972(3)
0.965(2)
0.964(1)
0.9686(7)
0.9635(8)
0.9651(7)
0.9539(7)
0.956(8)
0.966(4)
0.9696(8)
0.9631(8)
0.955(1)
0.9629(9)
0.95(1)
0.962(2)
0.955(1)
0.956(1)
0.9695(9
0.9644(8)
0.957(1)
0.92(1)
0.959(1)
0.955(1)
0.9660(8)
0.956(1)
0.9644(8)
0.9594(9)
0.9618(8)
0.965(1)
0.95(2)
0.948(8)
0.962(1)
0.9593(9)
0.9653(8)
0.961(1)
0.9669(8)
0.944(3)
0.957(2)
0.953(1)
0.9569(8)
0.9589(9)
0.957(1)
0.962(1)
0.936(3)
0.9528(9)
0.949(1)
0.9515(9)
0.9384(8)
0.9457(9
0.9512(9)
0.952(1)
0.869(2)
0.895(1)
0.890(1)
0.907(1)
4.5
4.0
3.5
3.0
2.5
2.0
η
10^1
10^2
p [GeV/c]
1.00
0.98
0.96
0.94
0.92
0.90
0.88

2D binning



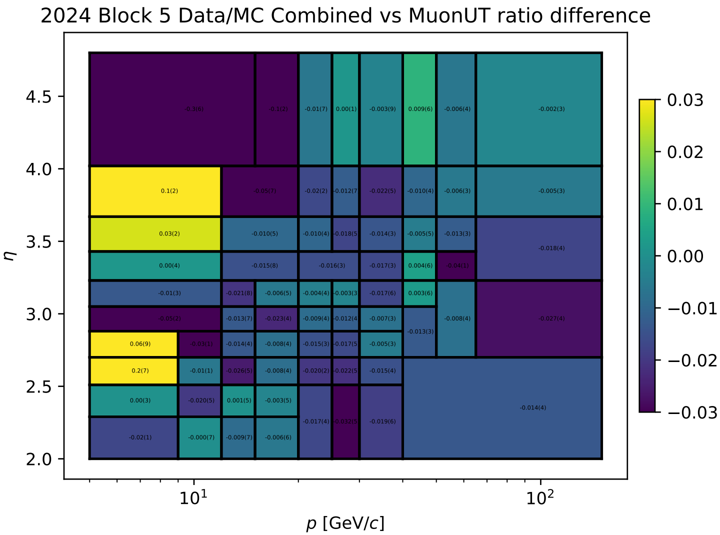
2024 Block 5 MC tracking efficiencies
η
p [GeV/c]
10^1
10^2
4.5
4.0
3.5
3.0
2.5
2.0
1.00
0.98
0.96
0.94
0.92
0.90
0.88

2D binning



2024 Block 5 Data/MC tracking efficiency ratios
0.7(4)
0.85(4)
0.98(1)
1.007(5)
0.989(3)
0.998(3)
0.987(2)
0.992(2)
1.06(9)
0.985(7)
0.984(3)
0.978(3)
0.982(2)
0.990(2)
0.989(2)
0.981(2)
1.01(1)
0.988(3)
0.984(3)
0.986(2)
0.988(1)
0.982(2)
0.988(2)
0.979(2)
0.981(9)
0.983(4)
0.985(1)
0.984(2)
0.977(2)
0.984(2)
0.98(1)
0.984(3)
0.982(2)
0.983(2)
0.988(2)
0.984(2)
0.982(3)
0.95(1)
0.987(3)
0.974(2)
0.979(2)
0.978(2)
0.982(2)
0.986(2)
0.980(3)
0.982(2)
0.99(2)
0.968(8)
0.986(3)
0.981(2)
0.982(2)
0.983(3)
0.989(2)
0.961(4)
0.976(4)
0.975(2)
0.977(2)
0.980(2)
0.976(3)
0.982(2)
0.972(4)
0.977(2)
0.975(2)
0.977(2)
0.981(3)
0.953(4)
0.956(3)
0.949(3)
0.957(3)
0.970(2)
0.971(2)
0.973(2)
η
4.5
4.0
3.5
3.0
2.5
2.0
10^1
10^2
p [GeV/c]
1.02
1.01
1.00
0.99
0.98
0.97
0.96
0.95

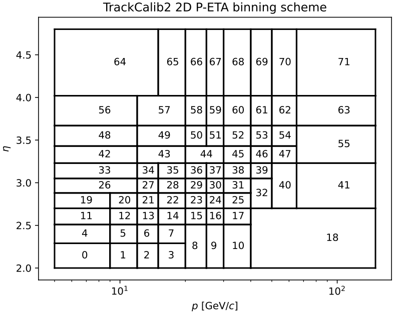
2D binning

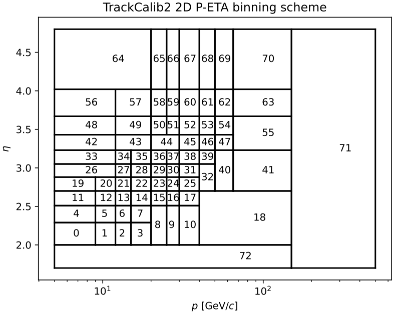


2024 Block 5 Data/MC Combined vs MuonUT ratio difference
η
4.5
4.0
3.5
3.0
2.5
2.0
10^1
10^2
p [GeV/c]
0.03
0.02
0.01
0.00
−0.01
−0.02
−0.03

Note: After some feedback, the 2D binning will be slightly modified

- Merge bins 64 and 65
- Add a single high p bin and a low η bin
  - Bins with low statistics, but could be useful for some analyses to have





Combined vs MuonUT systematic for 2024

Combined vs MuonUT systematic strategy for 2024?

- Clearly depends on kinematics, so we should use 2D binning
- It's a method systematic, should not depend on dataset with same μ
- Our current strategy:
  1. Determine difference in data/MC ratio for all data sets with 2D binning
  2. Do weighted average of the difference for blocks 5/6 and 7/8
  3. For bins with a discrepancy above $1\,\sigma$, assign the difference from zero as a systematic
  4. Use blocks 5/6 as proxy for blocks 1/2 due to bug in MuonUT trigger line (see !4047 and !4050)
  5. Systematic will be small in most bins, but a small number of bins have a 1%–2% systematic, in rare cases $> 3\%$

Release of tracking efficiencies

How should analysts use these results?

- Binning schemes are defined in JSON files
- Data/MC ratio in 2D bins provided for physics analysis, also as JSON
- Version 0 is available as a GitLab project here
- Version 1 will come soon
  1. Waiting for large MC production to finish
     - Current samples have insufficient statistics to properly constrain mass shapes
     - Statistical uncertainties are dominated by MC statistics in some bins
  2. Separate μ^+ and μ^- results as well
  3. Updated 2D binning scheme

Part II

Part I: Tracking efficiencies with $J/\psi \to \mu^+\mu^-$

Part II: Hadronic corrections

Introduction to hadronic corrections

- About 11% of kaons and 14% of pions cannot be reconstructed due to hadronic interactions with the detector material
- Modifies the track reconstruction efficiency of long muon tracks

$$\varepsilon_{hadron}^{reco} = \varepsilon_{muon}^{reco} \times \rho_{hadron}^{had.Int.}$$

- Simulation of hadronic interactions has uncertainties due to the material distribution, which is hard to quantify
- In Run 2: Assumed 10% uncertainty $\rightarrow$ 1.1% and 1.4% uncertainty on the kaon and pion tracking efficiency, respecively
- This work: Measure tracking efficiency correction factors $\rho_{hadron}^{had.Int.,Data}/\rho_{hadron}^{had.Int.,MC}$ from data

Method overview

- Measure the relative efficiency of $D^0 \to K^-\pi^+$ and $D^0 \to K^-\pi^+\pi^+\pi^-$ in data (blocks 1,2,5-8) and simulation (Sim10g)
- The total reconstruction&selection efficiency $\varepsilon^{reco\&sel}$ is factorised as

$$\varepsilon^{reco\&sel} = \varepsilon^{acc}\varepsilon^{reco}\varepsilon^{HLT1}\varepsilon^{HLT2}\varepsilon^{offline} \equiv \varepsilon^{reco}\varepsilon^{nuis}$$

- Can isolate hadronic correction for two pions $\rho_{\pi\pi}^{had.Int.}$

$$\frac{\varepsilon_{K\pi\pi\pi}^{reco\&sel}}{\varepsilon_{K\pi}^{reco\&sel}} = \frac{\varepsilon_{\mu\mu\mu\mu}^{reco}\varepsilon_{K\pi\pi\pi}^{nuis}\rho_{K\pi\pi\pi}^{had.Int}}{\varepsilon_{\mu\mu}^{reco}\varepsilon_{K\pi}^{nuis}\rho_{K\pi}^{had.Int}} = \frac{\varepsilon_{\mu\mu\mu\mu}^{reco}\varepsilon_{K\pi\pi\pi}^{nuis}}{\varepsilon_{\mu\mu}^{reco}\varepsilon_{K\pi}^{nuis}}\rho_{\pi\pi}^{had.Int}$$

- Use this information to infer correction for a single pion track $\rho_{\pi}^{had.Int}$

Measure $\rho_{\pi\pi}^{had.Int}$ in data and MC

- Measure $\rho_{\pi\pi}^{had.Int.}$ in data and MC

$$\rho_{\pi\pi}^{had.Int,Data} = \frac{N_{K\pi\pi\pi}^{fit}}{N_{K\pi}^{fit}} \frac{\mathcal{B}(D^0 \to K^-\pi^+)}{\mathcal{B}(D^0 \to K^-\pi^+\pi^+\pi^-)} \frac{\varepsilon_{\mu\mu}^{reco}}{\varepsilon_{\mu\mu\mu\mu}^{reco}} \frac{\varepsilon_{K\pi}^{nuis}}{\varepsilon_{K\pi\pi\pi}^{nuis}}$$

$$\rho_{\pi\pi}^{had.Int,MC} = \frac{N_{K\pi\pi\pi}^{sel}}{N_{K\pi\pi\pi}^{gen}} \frac{N_{K\pi}^{gen}}{N_{K\pi}^{sel}} \frac{\varepsilon_{\mu\mu}^{reco}}{\varepsilon_{\mu\mu\mu\mu}^{reco}} \frac{\varepsilon_{K\pi}^{nuis}}{\varepsilon_{K\pi\pi\pi}^{nuis}}$$

- Calculate data/MC correction $\rho_{\pi\pi}^{had.Int.,Data}/\rho_{\pi\pi}^{had.Int.,MC}$:
  - Assume $\frac{\varepsilon_{K\pi}^{nuis,Data}/\varepsilon_{K\pi\pi\pi}^{nuis,Data}}{\varepsilon_{K\pi}^{nuis,MC}/\varepsilon_{K\pi}^{nuis,MC}} = 1$
  - $\varepsilon_{\mu\mu}^{reco}/\varepsilon_{\mu\mu\mu\mu}^{reco}$ from TrackCalib2
  - $\mathcal{B}(D^0 \to K^-\pi^+\pi^+\pi^-)/\mathcal{B}(D^0 \to K^-\pi^+) = 2.083 \pm 0.031$ from PDG

Kinematic reweighting

- High-p_T π^+ in $D^0 \to K^-\pi^+\pi^+\pi^-$ has most similar kinematics to π^+ in $D^0 \to K^-\pi^+$
- For $K^-\pi^+_{(\mathrm{high}p_T)}$ to cancel, ensure cover same phase-space by reweighting:

1. Reweight $D^0 \to K^-\pi^+\pi^+\pi^-$ MC to sWeighted data in (p, η) for $K^-, \pi^+_{\mathrm{high}p_T}$ and ϕ for each track
2. Reweight sWeighted $D^0 \to K^-\pi^+$ data to sWeighted $D^0 \to K^-\pi^+\pi^+\pi^-$ data in (p, η) for $K^-, \pi^+_{(\mathrm{high}p_T)}$
3. Reweight $D^0 \to K^-\pi^+$ MC to sWeighted & kin. reweighted $D^0 \to K^-\pi^+$ data

Phase-space integrated hadronic correction

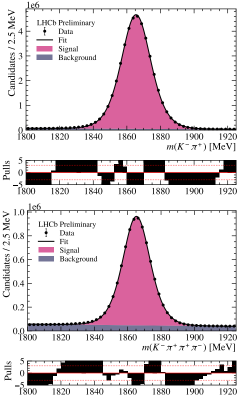
- Single fit to $D^0 \to K^-\pi^+$ and $D^0 \to K^-\pi^+\pi^+\pi^-$ data
- Kinematic corrections applied to $D^0 \to K^-\pi^+$ data

$$\rho_{\pi\pi}^{had.Int.,Data/MC} = 100.88 \pm 1.51\%$$

- Correction for single hadron track

$$\rho_{\pi}^{had.Int.,Data/MC} = 100.44 \pm 0.75\%$$

- Uncertainty dominated by $\mathcal{B}(D^0 \to K^-\pi^+\pi^+\pi^-)/\mathcal{B}(D^0 \to K^-\pi^+)$

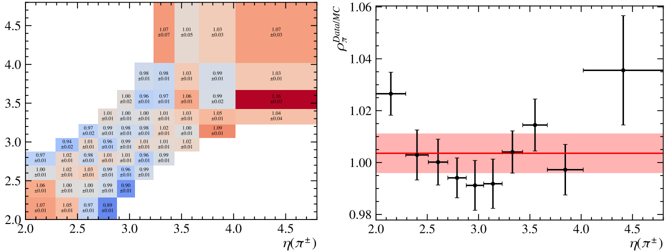


$D^0 \to K^-\pi^+$ (top) and $D^0 \to K^-\pi^+\pi^+\pi^-$ (bottom), 2024 Block 1

Hadronic correction as function of η

- Check for dependence of the hadronic correction factor on η
- Dependence on $K^-\pi^+_{(\text{high } p_T)}$ cancels in the ratio...
- ... but not $\pi^+_{\text{low } p_T}$ and π^- from $D^0 \to K^-\pi^+\pi^+\pi^-$
- $\to$ Need to bin in $\eta(\pi^+_{\text{low } p_T})$ and $\eta(\pi^-)$
  - Use same bin edges as muon tracking efficiency

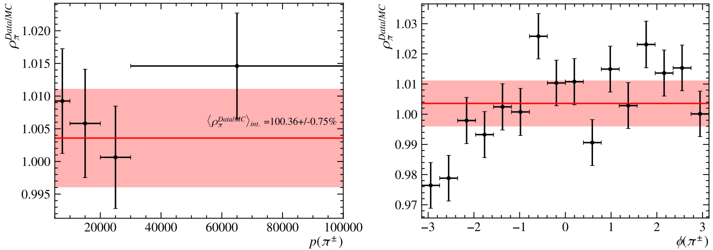
- Measure $\rho_{\pi\pi}^{had.Int}$ in bins of (η, η)
- Obtain 1D distribution by assuming factorisation $\rho_{\pi\pi}^{had.Int.} = \rho_{\pi}^{had.Int.}\rho_{\pi}^{had.Int.}$
- Note: Results are currently very preliminary, cross checks are ongoing



Hadronic correction factor for two pions (left) in 2d bins of η (left) and hadronic correction factor for a single pion (right).

Hadronic correction as function of p or ϕ

- Can also bin in (p, p) or (ϕ, ϕ) and measure the hadronic correction factor as function of p or ϕ
- As expected, no dependence on p
- Drop around $\phi \sim -\pi$ observed and is being investigated



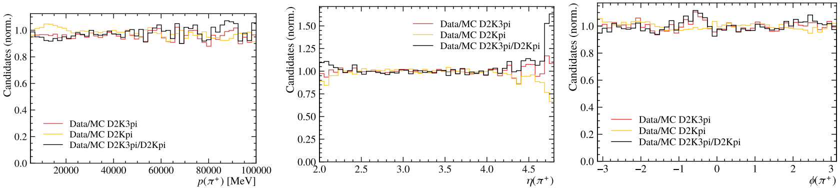
Hadronic correction factor for a single track in bins of p (left) and ϕ (right).

Cancellation of $K^-\pi^+_{(\text{high } p_T)}$

- So far, assume perfect cancellation between $K^-\pi^+$ from $D^0 \to K^-\pi^+$ and $K^-\pi^+_{\text{high } p_T}$ from $D^0 \to K^-\pi^+\pi^+\pi^-$
- However, some non-cancellation at the edges of p, η and ϕ seen

$\rightarrow$ Need to revisit the kinematic reweighting

- Structures seen as function η and ϕ might be (partially) due to this



Ratio of data/MC for $D^0 \to K^-\pi^+\pi^+\pi^-$ (red), $D^0 \to K^-\pi^+$ (yellow) and the double-ratio (black) of the π^+ cancelling between the two decays in bins of p (left), η (centre) and ϕ (right).

Summary and next steps

Tracking efficiencies with $J/\psi \to \mu^+\mu^-$

- Muon tracking efficiencies v0 for 2024 can now be used by analysts
- Issues to follow up on:
  - Block 1b MC
  - 2025 Velo efficiencies
- Waiting for large MC production before next release

Hadronic interactions

- Corrections determined from $D^0 \to K^-\pi^+$ and $D^0 \to K^-\pi^+\pi^+\pi^-$
- Well modelled in MC, and corrections provided in η bins
- Uncertainties are about 1% in most bins of η
- Still need to investigate
  - data/MC agreement in p, η and ϕ
  - cancellation of $K^-\pi^+$, nuisance effects

Thanks for your attention!

Backup

Overlap criteria in matching

Matching is done by comparing the hits of the probe track with a long track in each subdetector

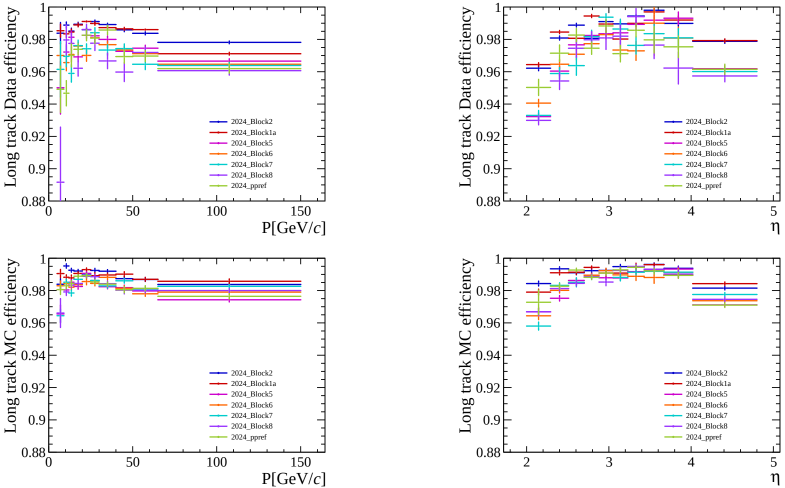
Method	Velo	UT	SciFi	Muon
VeloMuon	$> 40\%$	–	–	$> 40\%$
Downstream	–	–	$> 40\%$	$> 40\%$
MuonUT	–	0 or $> 40\%$	–	$> 40\%$

Note:

- Matching was originally performed in a trigger line
- We now have functors for direct access to overlap fractions
- Requires rerunning reconstruction in 2024, as long-track information was not persisted

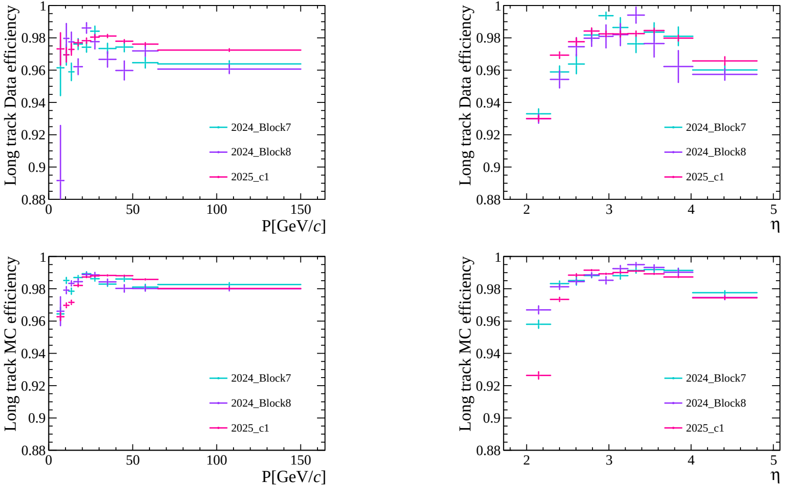
Efficiency comparisons 2024

Comparison of long tracking efficiencies in 2024 data blocks



Efficiency comparisons 2025

Comparison of long tracking efficiencies in 2025, compared with 2024

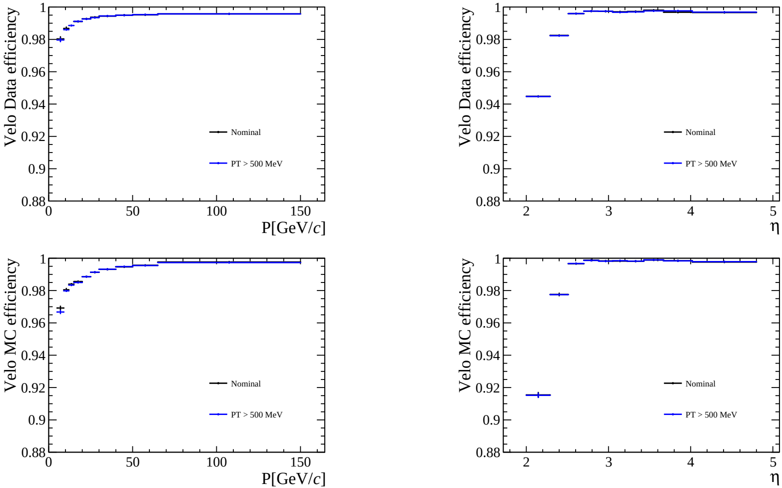


Changes to 2025 selection

- Probe muon p_T cut was at 500 MeV in 2024
- Loosened to 100 MeV in 2025
- Differences are small, but cannot compare 2024 and 2025 directly
- Comparison in backup

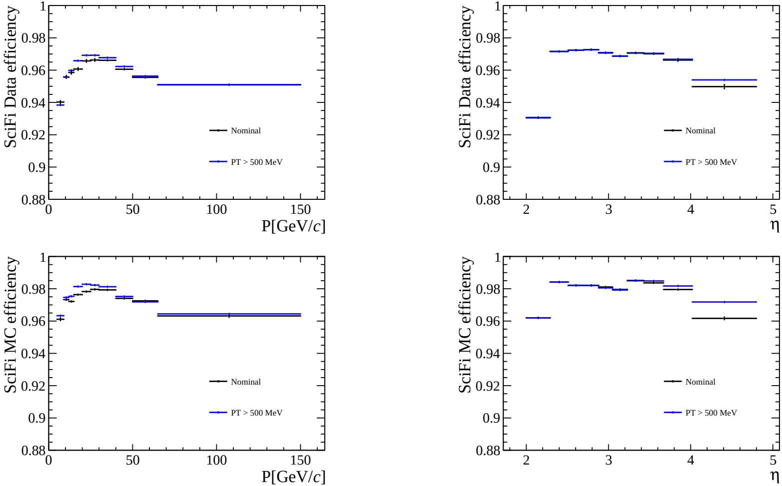
p_T cut comparison

Comparison of Velo tracking efficiencies with $p_T > 500\,\mathrm{MeV}$ cut



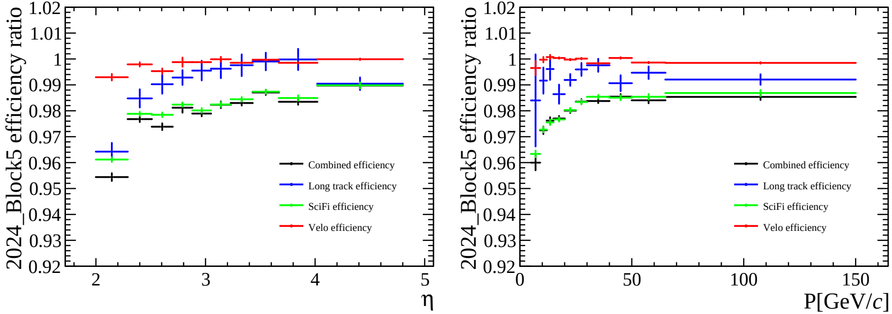
p_T cut comparison

Comparison of SciFi tracking efficiencies with $p_T > 500\,\text{MeV}$ cut



Tracking efficiency data/MC ratio

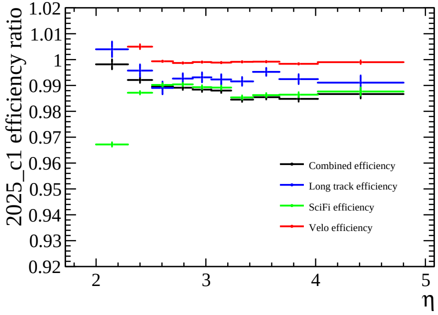
Data/MC ratio for Combined/MuonUT track efficiencies block 5

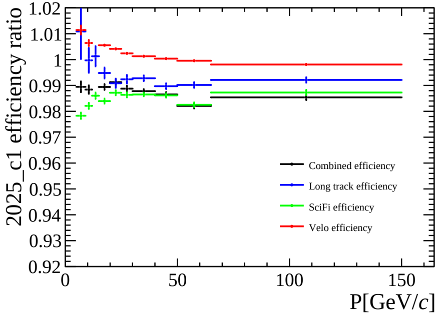


1. Very similar trends across 2024 data blocks
2. Combined efficiency is mostly driven by SciFi efficiency
3. Generally there is a 1–2% discrepancy between Combined and MuonUT efficiencies
   - Effect is larger at low p and low η, which must be understood

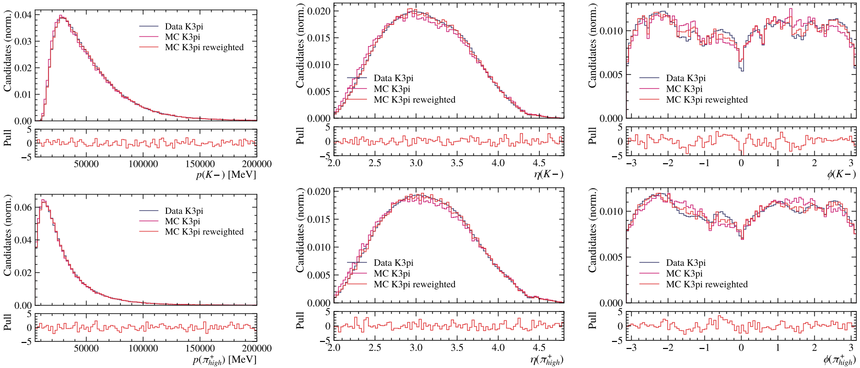
Tracking efficiency data/MC ratio

Data/MC ratio for Combined/MuonUT track efficiencies 2025 pre-TS



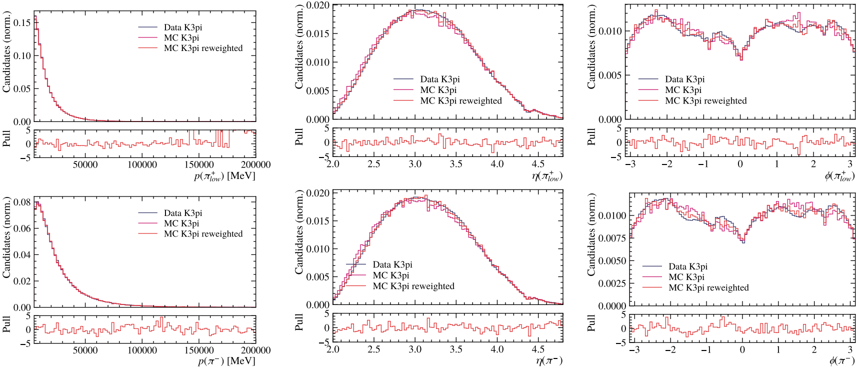


Data/MC comparison for $D^0 \to K^-\pi^+\pi^+\pi^-$



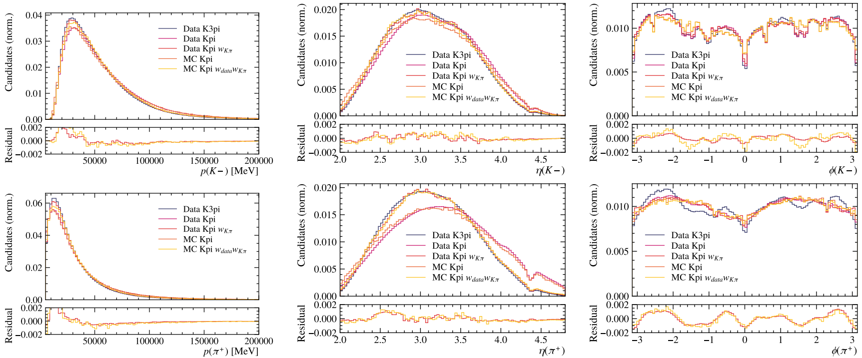
Comparison of data and simulation of $D^0 \to K^-\pi^+\pi^+\pi^-$ for K^- (top) and $\pi^+_{\text{high } p_T}$ (bottom) in p (left) and η (centre) and ϕ (right).

Data/MC comparison for $D^0 \to K^-\pi^+\pi^+\pi^-$



Comparison of data and simulation of $D^0 \to K^-\pi^+\pi^+\pi^-$ for $\pi^+_{\text{low } p_T}$ (top) and π^- (bottom) in p (left) and η (centre) and ϕ (right).

Data/MC comparison for $D^0 \to K^- \pi^+$



Comparison of data and simulation of $D^0 \to K\pi$ and $D^0 \to K\pi\pi\pi$ for K^- (top) and $\pi^+/\pi^+_{\text{high } p_T}$ (bottom) in p (left) and η (centre) and ϕ (right).

Uncertainties for phase-space integrated result

Source	$\Delta\rho_{\pi}^{had.Int.,Data/MC}$
$D \to K\pi$ simulation sample	0.04%
$D \to K3\pi$ simulation sample	0.07%
$D \to K\pi$ Data sample	0.01%
$D \to K3\pi$ Data sample	0.01%
Tracking weights	0.12%
$\mathcal{B}(D^0 \to K^-\pi^+\pi^+\pi^-)/\mathcal{B}(D^0 \to K^-\pi^+)$	0.74%
HLT selection	x.xx
Offline selection	x.xx
Total	0.76%

- Dominated by uncertainty on the branching ratio
- Uncertainty due to tracking weights from bootstrapping
- Expect HLT and offline uncertainties to cancel in the ratio